

TEST ITEM ANALYSIS REPORT

ACT 25MC1 - MATHEMATICS



Test Code: 25MC1

School: Sample High School

Grade: 11

Test Date: 10/31/2025

N* = 129

Number of Items = 41

Group: All Students Tested

ITEM	STRAND	BENCHMARK CODE	PERCENT CORRECT			A/F	B/G	C/H	D/J	E/K	BLANK	KEY	COMMON CORE**
			0%	50%	100%								
NBO Number and Quantity: Basic Operations & Applications													
32	411	Solve routine two or three-step arithmetic problems				5%	23%	9%	62%	0%	1%	J	
10	421	Solve two or three-step conversion problems within or between measurement systems				8%	18%	62%	12%	0%	1%	H	
22	421	Solve two or three-step conversion problems within or between measurement systems				6%	74%	15%	5%	0%	0%	G	
Strand Percent Average Correct for 3 items = 66%													
NNC Number and Quantity: Numbers: Concepts & Properties													
41	511	Order fractions				5%	23%	9%	61%	0%	2%	D	
8	551	Work with problems involving positive integer exponents, including squares/square roots and cubes/cube roots				7%	17%	65%	10%	0%	1%	H	
18	571	Exhibit knowledge of the basics of complex number addition and subtraction				4%	24%	9%	63%	0%	1%	J	
6	681	Use addition, subtraction, and scalar multiplication of vectors and of matrices				6%	24%	7%	62%	0%	1%	J	
Strand Percent Average Correct for 4 items = 63%													
AXE Algebra: Expressions, Equations, & Inequalities													
9	421	Solve routine first-degree equations				6%	23%	10%	60%	0%	1%	D	
17	431	Add and subtract simple algebraic expressions				55%	22%	15%	8%	0%	0%	A	
5	451	Multiply two binomials				18%	20%	36%	22%	0%	4%	C	
33	521	Solve real-world problems by using first-degree equations				8%	16%	65%	10%	0%	1%	C	
15	531	Add, subtract, multiply and divide polynomials				57%	22%	13%	9%	0%	0%	A	
14	621	Solve systems of two linear equations				5%	26%	9%	60%	0%	1%	J	
26	631	Manipulate expressions and equations				6%	71%	16%	7%	0%	0%	G	
37	652	Factor polynomial equations to find roots or zeros				29%	9%	57%	5%	0%	0%	C	
Strand Percent Average Correct for 8 items = 57%													
FFU Functions: Functions													

Items 7, 16, 29 and 40 are field-tested and not included in the scoring.

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				0%	50%	100%								
11	411	Evaluate linear and quadratic functions, expressed in function notation, in integers					6%	71%	16%	8%	0%	0%	B	
19	461	Write a missing term in a given pattern that has a constant factor between terms					6%	74%	15%	5%	0%	0%	B	
23	511	Evaluate polynomial functions, expressed in function notation, in integers					9%	19%	61%	11%	0%	1%	C	
13	551	Express the sine/cosine/tangent of an angle in a right triangle as a ratio of side lengths					57%	22%	13%	8%	0%	0%	A	
34	561	Find the next term in a sequence described recursively					6%	74%	15%	5%	0%	0%	G	
44	611	Evaluate composite functions at integer values					6%	24%	9%	60%	0%	0%	J	

Strand Percent Average Correct for 6 items = 66%

GGR Geometry: Graphical Representations

35	311	Locate both positive and negative rational numbers on the number line					26%	12%	57%	5%	0%	0%	C	
30	431	Locate points in the coordinate plane					21%	12%	12%	55%	0%	1%	J	
25	673	Given an equation or function, find an equation or function whose graph is a translation by a specified amount up or down					57%	20%	14%	9%	0%	0%	A	
24	681	Recognize special characteristics of parabolas and circles					21%	27%	26%	20%	0%	6%	G	

Strand Percent Average Correct for 4 items = 49%

GPP Geometry: Properties of Plane Figures

4	421	Exhibit knowledge of basic angle properties and special sums of angle measures					57%	22%	14%	8%	0%	0%	F	
21	541	Use Pythagorean triples to find the third length of a right triangle					26%	12%	58%	4%	0%	0%	C	
31	651	Apply properties of 30°-60°-90°, 45°-45°-90°, similar, and congruent triangles					5%	22%	60%	12%	0%	1%	C	

Strand Percent Average Correct for 3 items = 58%

GME Geometry: Measurement

20	221	Estimate or calculate the length of a line segment based on other lengths in a geometric figure or same direction line segments					9%	18%	62%	11%	0%	1%	H	
28	521	Compute the perimeter of simple composite geometric figures with unknown side lengths					57%	21%	14%	8%	0%	0%	F	

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			0%	50%	100%								
43	611	Use relationships involving area, perimeter, and volume to compute another measure				27%	9%	58%	5%	0%	0%	C	

Strand Percent Average Correct for 3 items = 59%

SPS Statistics/Probability: Probability, Statistics, & Data Analysis														
1	411	Calculate the missing data value given the average and all data values but one					6%	23%	10%	60%	0%	1%	D	
3	431	Determine the probability of a simple event					6%	71%	14%	9%	0%	0%	B	
38	511	Calculate the average given the frequency counts of all the data values					5%	25%	9%	60%	0%	1%	J	
39	511	Calculate the average given the frequency counts of all the data values					8%	18%	64%	10%	0%	1%	C	
12	531	Compute straightforward probabilities for common situations					7%	24%	9%	60%	0%	0%	J	
27	531	Compute straightforward probabilities for common situations					6%	74%	14%	6%	0%	0%	B	
36	531	Compute straightforward probabilities for common situations					6%	24%	9%	61%	0%	0%	J	
2	551	Use Venn diagrams in counting					7%	23%	9%	60%	0%	1%	J	
42	612	Exhibit knowledge of, calculate, or use the standard deviation					57%	22%	14%	7%	0%	0%	F	
45	631	Compute a probability when the event and/or sample space are not given or obvious					57%	22%	14%	7%	0%	0%	A	

Strand Percent Average Correct for 10 items = 62%

Overall Percent Average Correct for 41 items = 60%

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